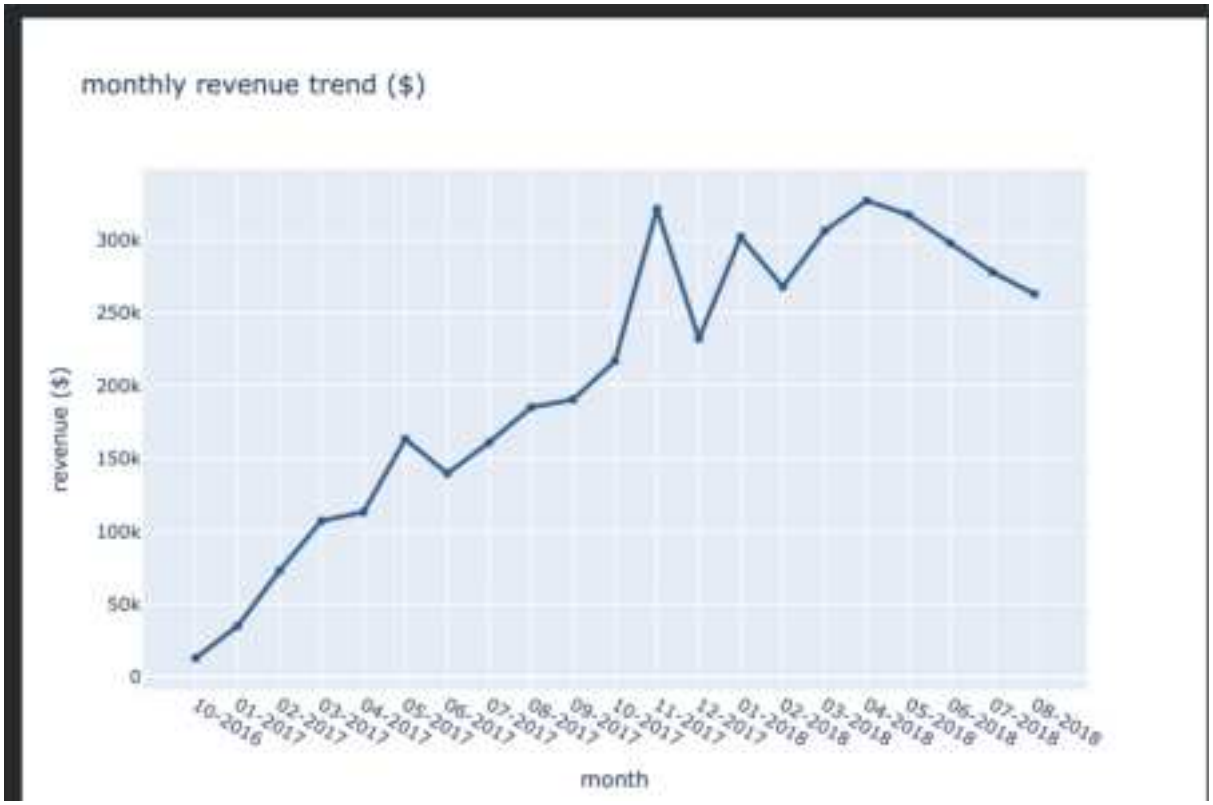
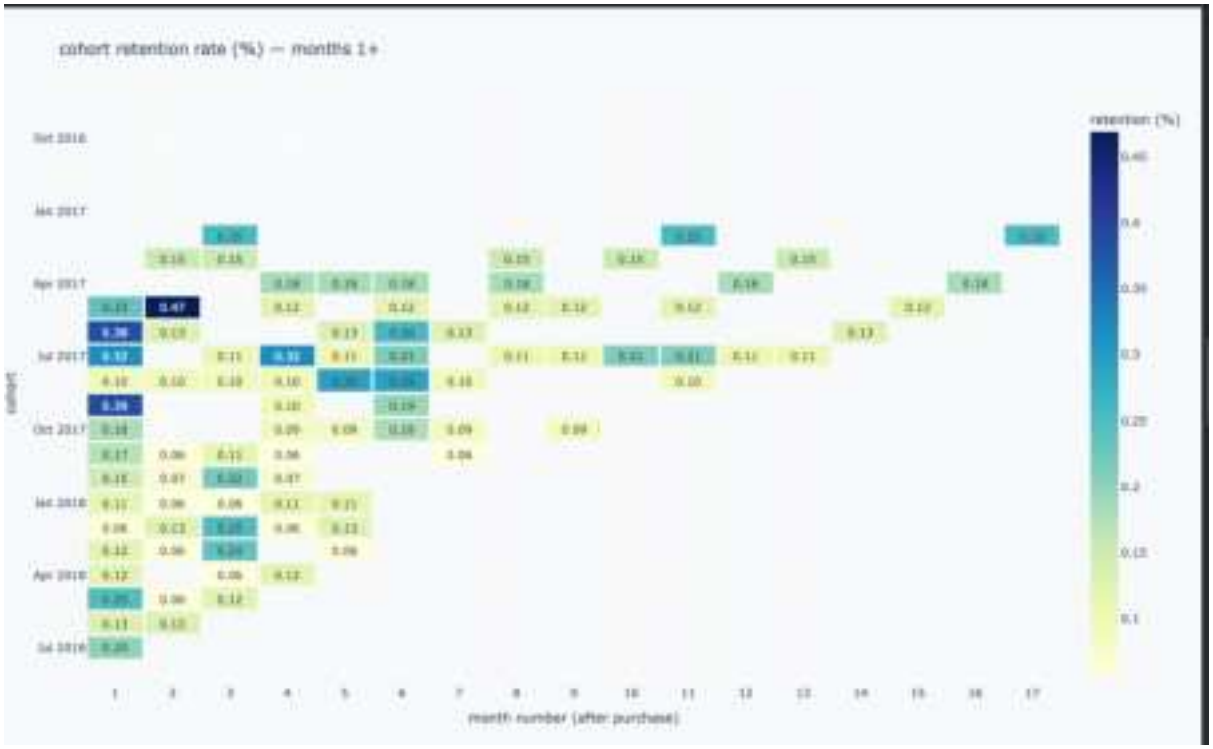


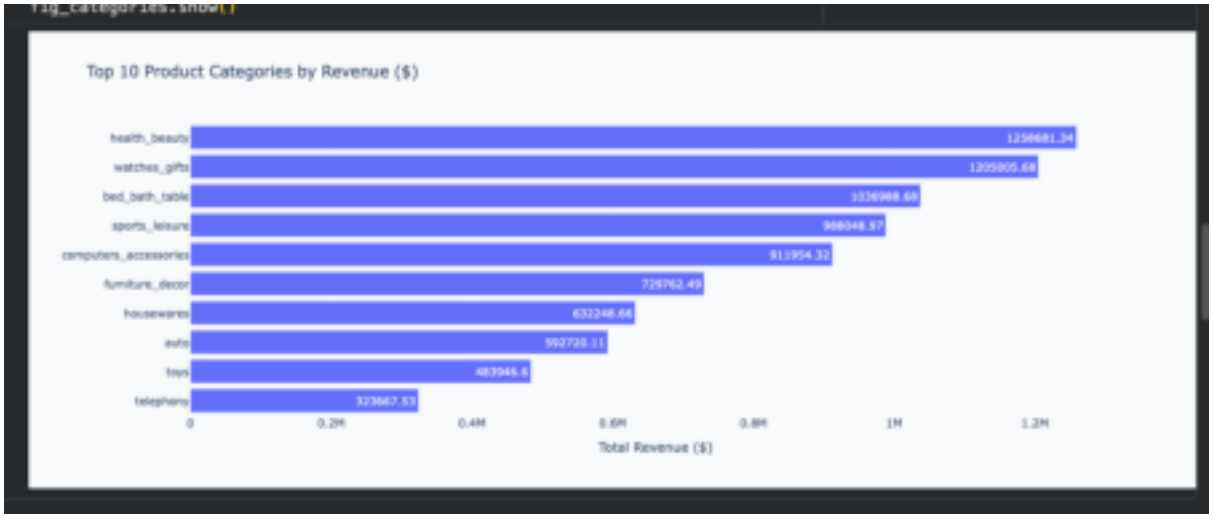
Fact: Mounthly reveue grew steadily from late 2016, reaching a peak og around 326k in March 2018, followed by a downward trend to 262k
Root Cause: Customers might be losing interest over time, or the platform lacks engagement triggers for repeat purchases.
Recommendation: We should introduce special offers, promotional campaigns, or a loyalty/bonus point system to re-engage users. Also, it is necessary to check which specific product categories experienced the highest demand drop



Fact: The cohort retention heatmap shows that repeat purchase rates are extremely low. The highest spikes appear in early cohorts (like Apr 2017), but for the vast majority of months, retention drops close to 0%.
Root Cause: The platform functions as a one-time purchase marketplace. Once customers buy an item (often large goods like furniture), they have no reason to return.
Recommendation: Since users do not return organically, the business needs automated email retention campaigns and push notifications right after the first delivery to encourage a second purchase within the first 30 days.

	category_english	total_orders	total_revenue	avg_freight_cost	freight_to_price_pct
0	bed_bath_table	9417	1036988.68	18.42	28.4
1	health_beauty	8836	1258681.34	18.88	30.2
2	sports_leisure	7720	988048.97	19.51	29.4
3	computers_accessories	6689	911954.32	18.82	29.4
4	furniture_decor	6449	729762.49	20.73	34.7
5	housewares	5884	632248.66	20.99	40.3
6	watches_gifts	5624	1205005.68	16.78	17.1
7	telephony	4199	323667.53	15.67	50.6
8	auto	3897	592720.11	21.88	33.2
9	toys	3886	483946.60	18.81	26.5

```
import plotly.express as px
```



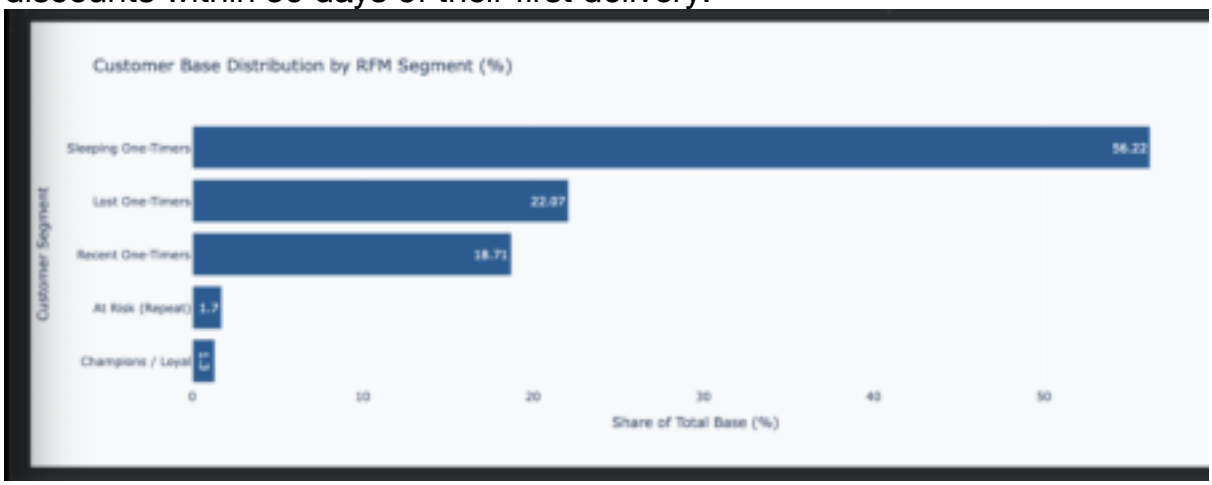
Fact: Revenue is led by health_beauty (1.25M) and watches_gifts (1.20M). However, bed_bath_table ranks highest in total order volume (9,417 orders). Lower-ranking categories like telephony show a very high freight-to-price ratio (50.6%).
Root Cause: High-ticket items (watches) drive revenue through higher average order value, while lower-priced categories rely on volume. Categories like telephony underperform because shipping costs account for over half of the product price, discouraging buyers.
Recommendation: Introduce bundled shipping discounts for low-cost categories (like phone accessories) to offset shipping fees and boost order volume.



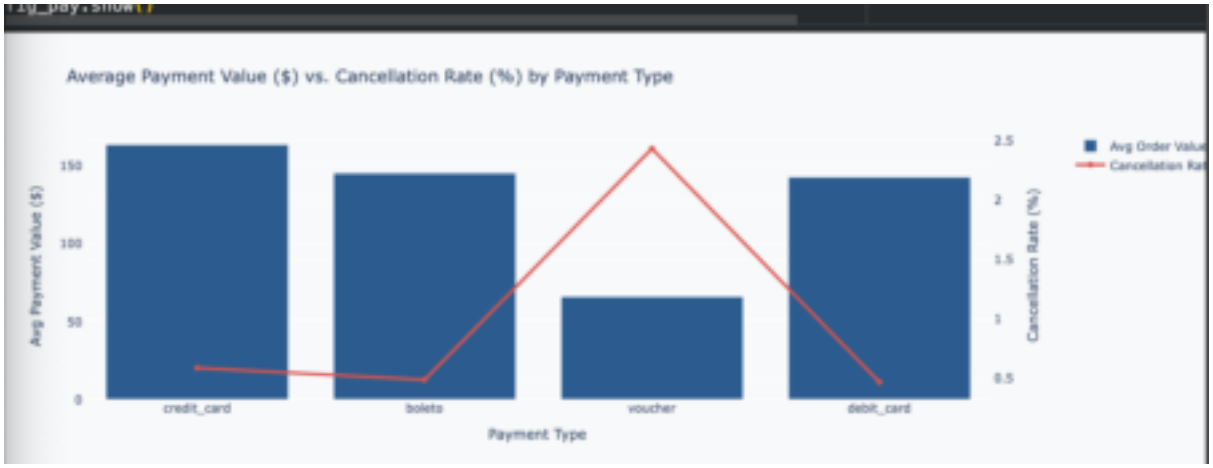
Fact: On-time deliveries receive an average customer satisfaction score of 4.29 out of 5, whereas delayed deliveries suffer a sharp drop to 2.57.
Root Cause: Delivery speed directly dictates customer perception and post-purchase satisfaction.
Recommendation: Set realistic estimated delivery dates and establish an automated notification system for logistically complex orders to mitigate negative reviews caused by unexpected delays.



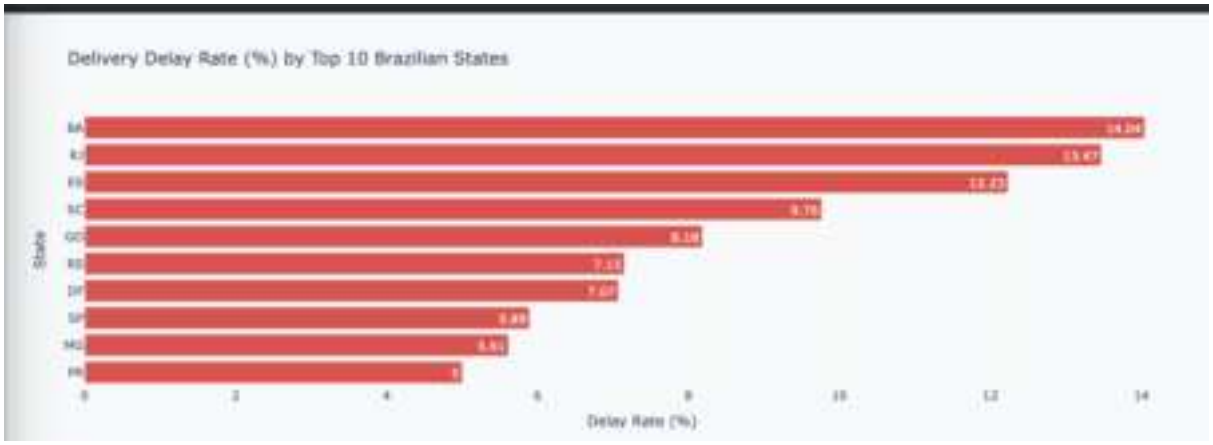
Fact: The repeat purchase rate stays extremely low (between 3.75% and 4.98%) across all review score ratings from 1 to 5 stars.
Root Cause: High customer satisfaction on the first purchase does not automatically drive loyalty. The platform lacks trigger loops and incentives to bring happy buyers back.
Recommendation: Focus retention efforts not only on resolving bad support experiences but also on actively marketing to 5-star buyers via trigger discounts within 30 days of their first delivery.



Fact: Over 97% of the customer base consists of one-time buyers (56.2% sleeping, 22.1% lost, 18.7% recent), while repeat segments (At Risk and Champions) make up less than 3% combined.
Root Cause: The business heavily relies on aggressive new user acquisition without an efficient CRM system to nurture repeat demand.
Recommendation: Build lifecycle email flows specifically for the Recent One-Timers (18.7%) segment during their first 60 days to prevent them from slipping into Sleeping status.



act: Credit cards and boleto account for the highest average order value (140-160) with cancellation rates below 0.6%. Vouchers show a lower average spend ($65) and a significantly higher cancellation rate (2.4%).
Root Cause: Vouchers are often associated with promotional testing or partial refunds, leading to higher order abandonment, whereas credit card users show higher purchase intent.
Recommendation: Audit the voucher checkout flow to eliminate technical bugs and limit low-value promotional vouchers that lead to unpaid cancellations



Fact: Delivery delay rates vary heavily by region, peaking in Bahia (BA - 14.0%) and Rio de Janeiro (RJ - 13.5%), compared to just 5.0%–5.9% in PR, MG, and SP. Root Cause: Logistics infrastructure and seller distribution are concentrated in the Southeast (São Paulo), causing severe shipping delays to northern and coastal states. Recommendation: Work with regional logistics partners in BA and RJ to optimize last-mile delivery and adjust estimated delivery timeframes for remote regions to avoid misleading expectations.

Summary

1. **High Revenue Concentration & Customer Churn (RFM & Retention):**

- Fact: Over 97% of active users are single-purchase buyers (Sleeping, Lost, or Recent One-Timers), while repeat customers account for less than 3% of the base.
- Impact: The platform operates as an acquisition-heavy marketplace with low Customer Lifetime Value (LTV), forcing continuous marketing expenditure to maintain GMV.

2. **Delivery Speed Directly Drives Customer Satisfaction:**

- Fact:Average review scores drop from 4.29 / 5 for on-time deliveries down to 2.57 / 5 for delayed orders.
- Impact:Logistic delays directly cause customer dissatisfaction and negative reviews, undermining word-of-mouth growth and organic retention.

3. **Geographical Logistics Bottlenecks:**

- Fact: Outer states like Bahia (BA - 14.0%) and Rio de Janeiro (RJ - 13.5%) experience significantly higher delay rates compared to core hubs like São Paulo (SP - 5.9%).
- Impact: Cross-country fulfillment issues create localized customer churn in remote markets.

4. Category Economics & Shipping Cost Friction:

- Fact: High-ticket categories like watches_gifts and health_beauty drive major GMV, whereas lower-priced categories like telephony face severe shipping friction, with freight costs reaching 50.6% of the product price.
- Impact: High relative delivery fees reduce purchasing conversion on budget items.

Solution

Implement Post-Purchase Retention Loops: Set up automated lifecycle email flows and discount triggers targeting Recent One-Timers within the first 30–60 days to improve repeat purchase rates. **Optimize Logistics in High-Delay Regions:** Partner with regional 3PL logistics hubs in BA and RJ to cut down

delivery times and adjust estimated delivery dates to manage user expectations. **Incentivize High-LTV & Low-Freight Categories:** Offer bundled shipping discounts on lower-priced accessories to decrease relative delivery cost barriers and increase total basket size (AOV)